

Empirical properties of directed acyclic graphs occurring in natural languages.

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Research Question

Do dependency structures in natural languages exhibit systematic structural constraints when compared to randomly generated trees of the same size?

Hypothesis

Natural language dependency trees follow structural constraints that make them different from randomly generated trees. They tend to have bounded branching (low arity), moderate depth, and lower structural density and variability. These constraints may arise from cognitive and communicative pressures that favor efficient sentence processing.

Methodology

The analysis will use dependency treebanks from the **SUD (Surface-Syntactic Universal Dependencies)** corpus.

For each sentence in the dataset:

1. Extract the dependency tree and represent it as a directed graph.
2. Measure tree depth, node arity, and graph density for each tree.
3. Generate a random tree with the same number of nodes as the original sentence tree.
4. Compute the same metrics and compare their distributions using statistical summaries and visualizations.

Sample Languages

To ensure cross-linguistic coverage, the study will analyze treebanks from the following languages in the SUD corpus:

English, Hindi, Spanish, German, French, Russian, Chinese, Arabic, Turkish, Japanese.